Edge AI - Principles and Practices

Module - Introduction

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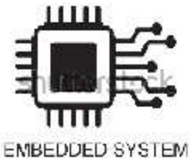
Module Objectives

- This module explains what edge AI is and its related terminologies.

Key Terms

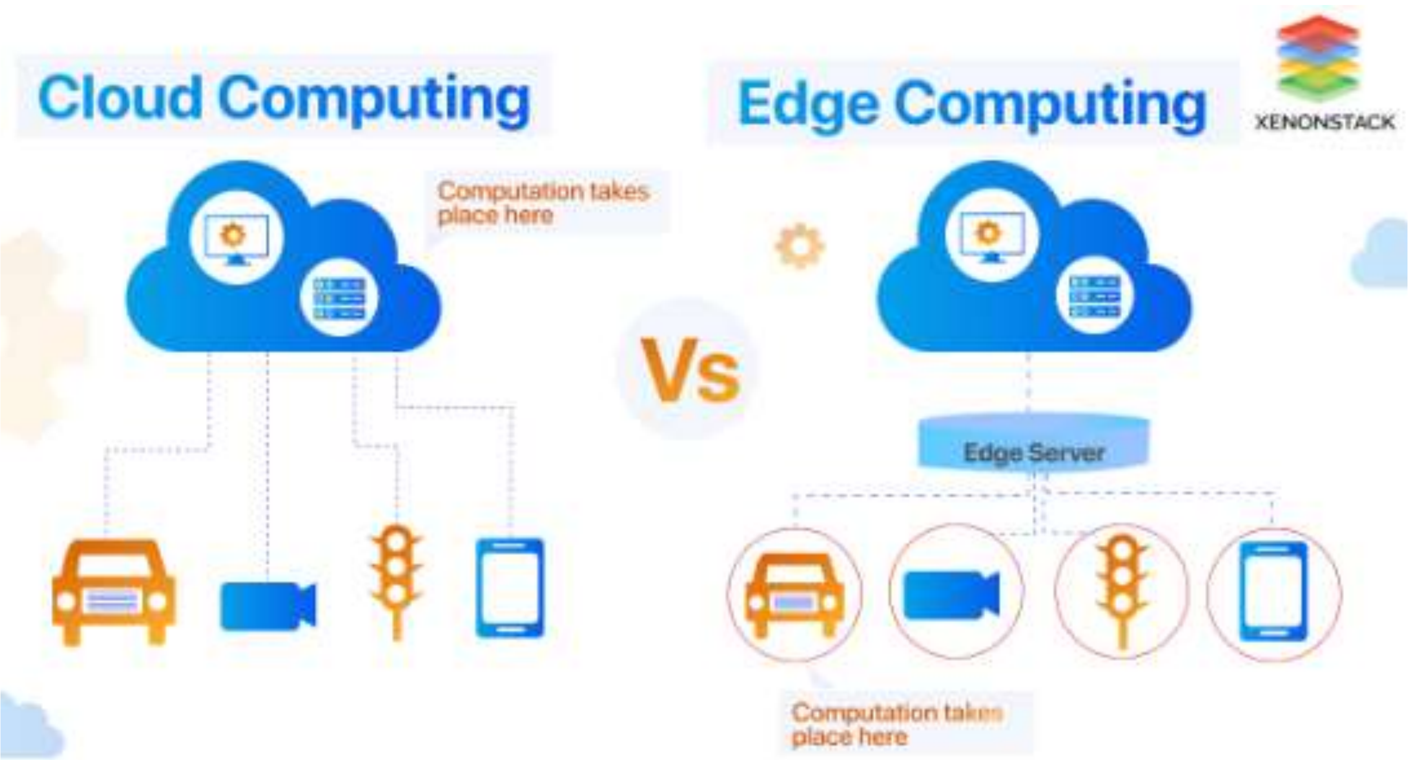
- Embedded systems (microcontrollers/microprocessors)



- Internet of Things

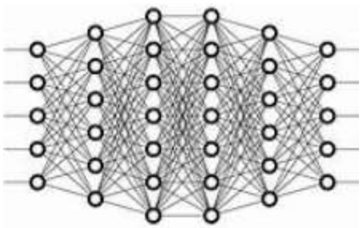
Key terms (continue)

- Edge vs. Cloud



Key terms (continue)

- AI (Artificial Intelligence / Neural Networks)



Key terms (continue)

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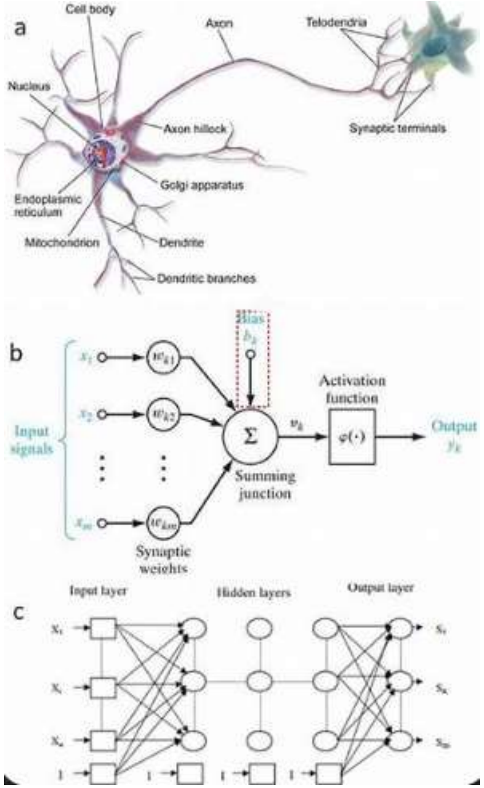
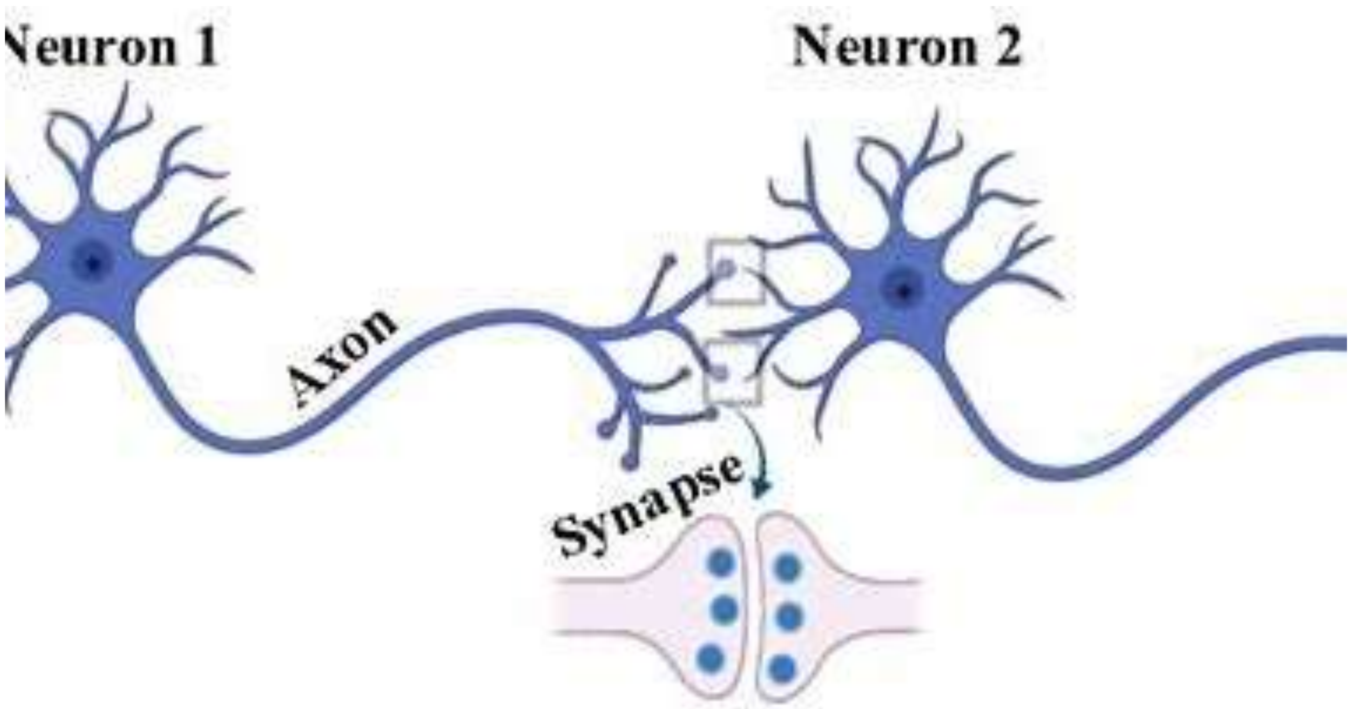
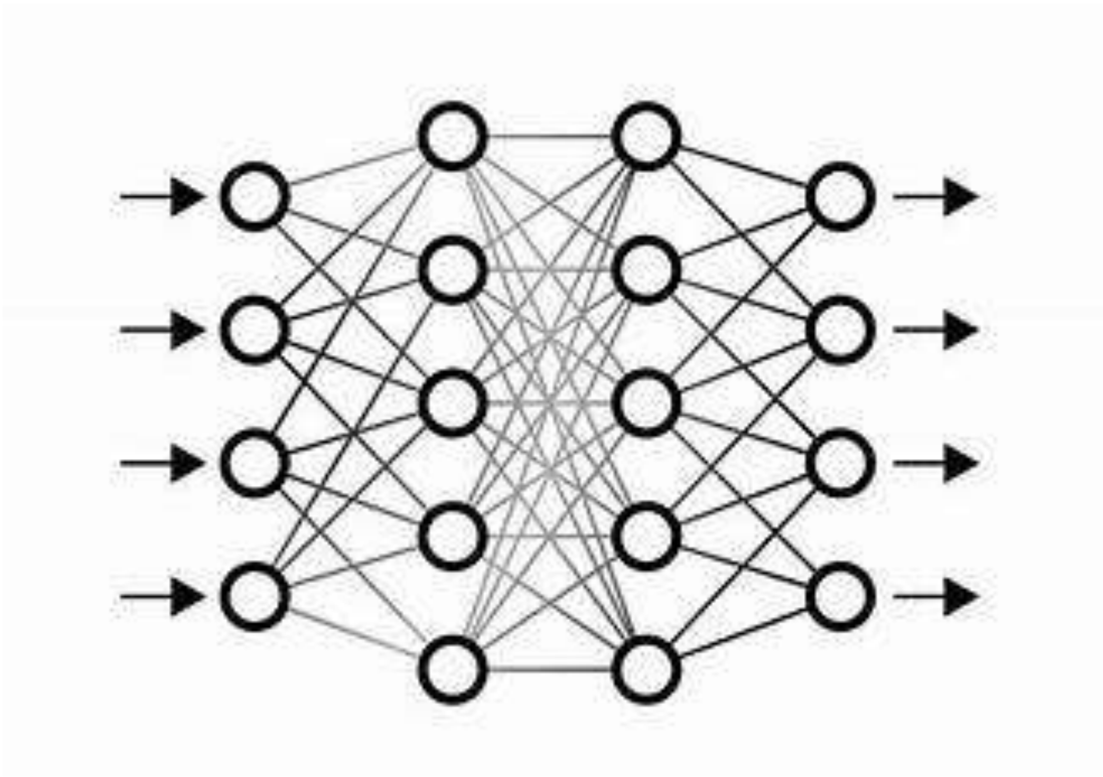


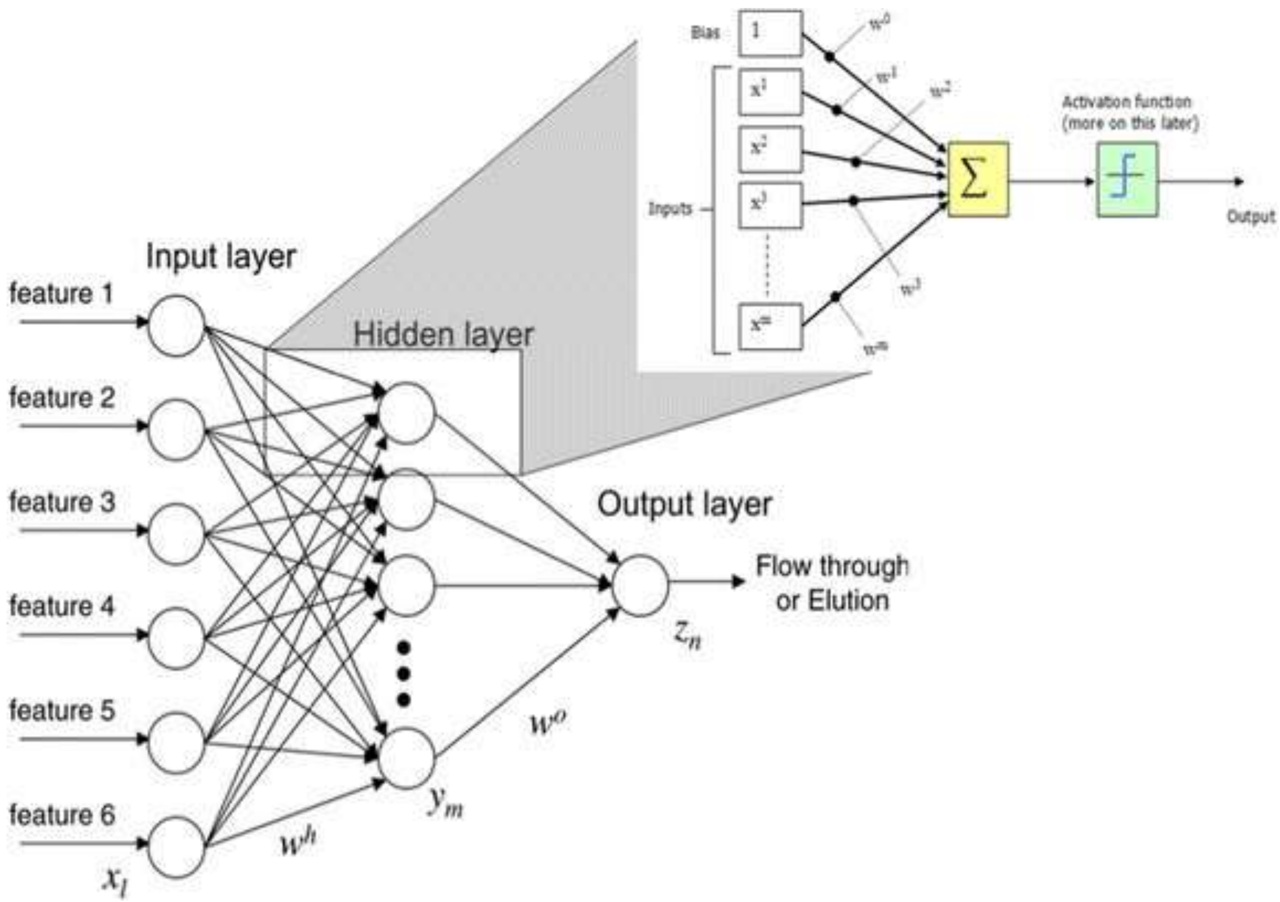
Key terms (continue)



Key terms (continue)



Key terms (continue)



Key terms (continue)

- ML (Machine Learning) – It is a broad and a bit vague term. In our course, we only focus on neural network learning/training.
- Deep Learning – Again, a bit vague and broad. In our course, we don't use this fancy term that much.
- Edge AI – Edge (local computation) + AI (neural networks – inference only)

Why Edge AI?

- In some applications we prefer Edge AI (of course, in other applications, we may prefer Cloud AI).
- Reasons:
  - Bandwidth
  - Latency
  - Economics
  - Reliability
  - Privacy

My Closing Thoughts

- ROI - Good enough (good Return on Investment)
- Good enough career (vs. cloud AI)
- Cheap playground - affordable, accessible (vs. what is your chance of tweaking Tesla's FSD or OpenAI's ChatGPT?)